

WASP-43b

R-1491

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_210328 · created 2026-08-01T13:49:31+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459301.7587 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.185 +/- 0.035
Transit depth (Rp/Rs)^2	3.43 +/- 1.29 [%]
Orbital Inclination (inc)	80.15 +/- 1.82
Airmass coefficient 1 (a1)	0.981 +/- 0.021
Airmass coefficient 2 (a2)	0.012 +/- 0.031
Scatter in the residuals of the lightcurve fit is	2.11 %
Best Comparison Star	#3 - [424, 93]
Optimal Aperture	2.79
Optimal Annulus	15.99
Transit Duration (day)	0.044 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.185 ± 0.035 vs 0.1594 (deviation 0.73σ)
Duration fit vs published	0.044 d vs 0.0512 d (deviation 0.72σ)
Mid-time O–C	-3.9 min
Depth SNR	5.3
Residual scatter	2.11 %

Light curve

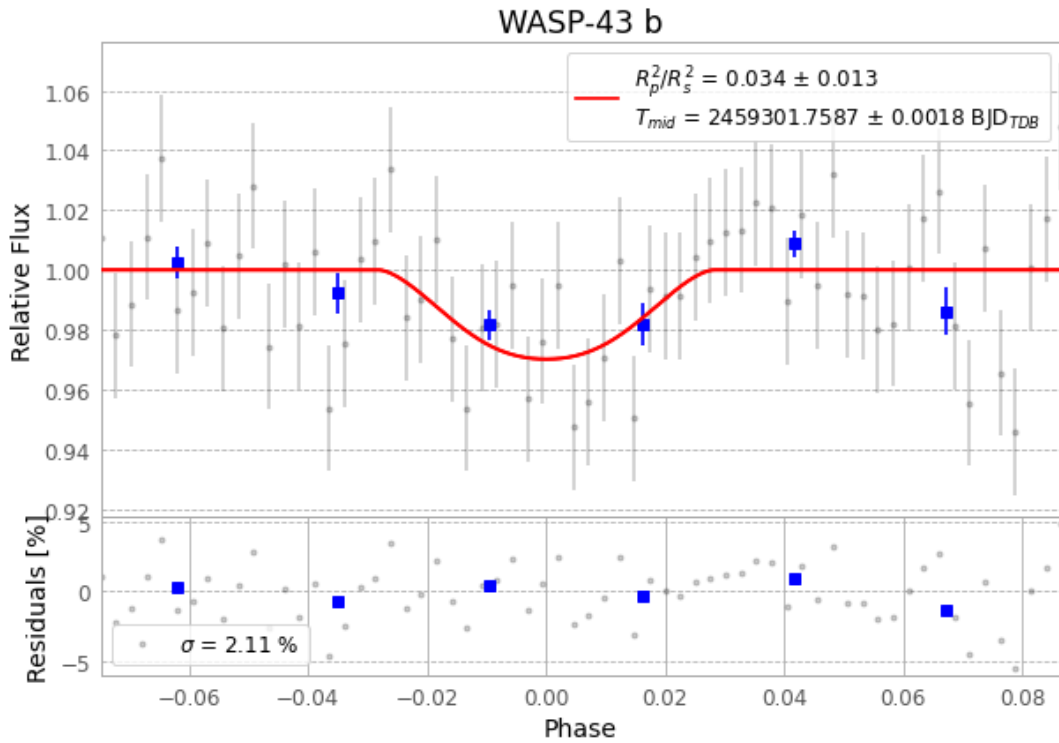


Figure 1 — FinalLightCurve_WASP-43 b_2021-03-27.png (SHA-256 25098e61e03a3cff...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [491, 96], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 6d01b2074ac3fbeb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

64 FITS frames (42 MB), WASP-43210328043612.FITS → WASP-43210328074512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-210328.log (d463d2563616...), FinalLightCurve_WASP-43 b_2021-03-27.png (25098e61e03a...), FinalLightCurve_WASP-43 b_2021-03-27.csv (fa1e1896dc1f...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 13:49Z.